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4 / 4 submitted

Level 1 — 100 marks/question

L1-Q1. leap year (E)

c

CODE

```
#include <stdio.h>
int main(){
    int year;
    printf("Enter a year:\n");
    scanf("%d",&year);
    if(year%4==0 && year%100!=0 || year%400==0){
        printf("%d is a leap year\n",year);
    }
    else{
        printf("%d is not a leap year\n",year);
    }
    return 0;
}
```

INPUT

2020

OUTPUT

Enter a year:
2020 is a leap year

L1-Q2. divisors (E)

c

CODE

```
#include <stdio.h>
int main(){
    int n,i,t=0;
    printf("Enter the number to get the divisors\n");
    scanf("%d",&n);
    for(i=1;i<=n;i++){
        if(n%i==0){
            t++;
        }
    }
    printf("Number of divisors:%d\n",t);
    return 0;
}
```

INPUT

12

OUTPUT

Enter the number to get the divisors
Number of divisors:6

L1-Q3. Armstrong (E)

c

CODE

```
#include <stdio.h>
int main(){
    int low,high,i,temp,num,digit,sum;
    int first = 1;
    printf("Enter the lower limit:\n");
    scanf("%d",&low);
    printf("Enter the upper limit:\n");
    scanf("%d",&high);
    printf("Armstrong numbers between %d and %d are:",low,high);
    for(i=low;i≤high;i++){
        int count=0;
        temp=i;
        while(temp≠0){
            count++;
            temp/=10;
        }
        sum=0;
        temp=i;
        while(temp≠0){
            digit=temp%10;
            int p=1;
            for(int j=0;j<count;j++){
                p*=digit;
            }
            sum+=p;
            temp/=10;
        }
        if(sum==i){
            if(!first){
                printf(", ");
            }
            printf("%d",i);
            first=0;
        }
    }
    printf("\n");
    return 0;
}
```

INPUT

100
500

OUTPUT

Enter the lower limit:
Enter the upper limit:
Armstrong numbers between 100 and 500 are:153, 370, 371, 407

L1-Q4. Square Pattern (E)

C

CODE

```
#include <stdio.h>
int main(){
    int n,i,j;
    printf("Enter the rows to print the square pattern:\n");
    scanf("%d",&n);
    for(i=1;i≤n;i++){
        for(j=1;j≤n;j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

INPUT

5

OUTPUT

```
Enter the rows to print the square pattern:
*****
*****
*****
*****
*****
```